Numerical simulation of 2D premixed combustion within a Rotating Detonation Engine

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Abstract

Rotating detonation engines (RDEs) emerge as a promising technology in the field of propulsion, potentially offering improved efficiency compared to traditional combustion engines. Nonetheless, the combustion process within RDEs currently lacks a comprehensive understanding. This study endeavors to bridge this knowledge gap by examining the flow behavior in a premixed combustion RDE model in detail. Utilizing a simplified 2D geometry, the research simulates the flow behavior of a hydrogen and air mixture in an RDE. The analysis of the computational domain occurs through the use of Ansys Fluent, with the combustion process modeled via a 10-species and 21-reaction mechanism. The study concentrates on the temporal evolution of the detonation wave, pressure and temperature distribution, shock wave interactions, flow characteristics, and detonation front dynamics. The findings offer insights into the RDE's flow behavior. Temporal analysis reveals stable detonation wave propagation, pointing to potential reliability and consistency of the engine. Observations indicate a notable increase in pressure and temperature distribution downstream of the detonation wave, signaling the release of thermal energy. Grasping the flow behavior of an RDE in terms of time and domain augments the current understanding of its performance, combustion characteristics, and potential areas for further exploration. Consequently, this study stands as a significant contribution towards fluidic understanding and addressing the hitherto limited data concerning RDEs, steering towards their successful incorporation in future propulsion systems.

Keywords: Rotating detonation engines,Detonation propulsion, Detonation wave, Pressure gain combustion, Advanced propulsion systems, Computational fluid dynamics

1 Introduction

Propulsion engines traditionally utilize deflagration, a subsonic and isobaric combustion process. As the quest for enhanced engine efficiency continues, several challenges become apparent, including stability regulation, weight reduction, combustion improvement, and the enhancement of material properties. This necessitates a reevaluation of engine designs to meet the continuously increasing performance demands of rockets.

Research indicates that detonation, a supersonic shock-driven combustion process, serves as a potential combustion mechanism for propulsion engines. This combustion is characterized by a shock wave that generates high downstream pressure and temperature [1], initiating a chemical reaction. The exothermic nature of this reaction generates heat sufficient to sustain a supersonic detonation structure. Detonation offers superior thermal efficiency and, as Raina [2] states, the reaction products have low entropy, making it more favorable than deflagration. However, incorporating this self-sustaining combustion method into modern systems poses challenges, primarily due to our limited understanding of the complex detonation phenomena. Existing models, often based on assumptions, do not fully capture the transient and three-dimensional nature of detonation [1].

Historically, Chapman, Jouguet, and Mikelson [3] introduced a foundational one-dimensional steady detonation theory, focusing on conservation laws in reacting mixtures. However, when their models failed to predict detonation velocity from conservation equations and the equation of state alone, Chapman and Jouguet formulated empirical criteria to determine this parameter [3]. Despite the lack of a theoretical basis, these criteria align well with subsequent studies and remain useful today due to their computational efficiency. The modern detonation theory, formulated by Zeldovich, von Neumann, and Doring [4], builds upon this early work by presenting the detonation structure as a combination of a shock wave, an induction zone, and a reaction zone.

Currently, two primary detonation combustors integrate into propulsion engines: Pulse Detonation Engines (PDEs) [5] and Rotational Detonation Engines (RDEs) [6]. PDEs operate through intermittent detonation in tubes, whereas RDEs establish a continuous detonation wave traveling around an annular chamber. Although innovative, PDEs encounter challenges including complex valving, repeated ignition, and a limited frequency range [7]. RDEs offer advantages over PDEs, encompassing continuous valveless injection, reduced weight, and adaptability to varying conditions [8]. Additionally, the RDE's simple annular structure facilitates integration, and its high-frequency operation minimizes unsteadiness [9]. Consequently, RDEs are becoming the preferred choice for future detonation engine applications [7].

Despite these benefits, a comprehensive understanding of the complex combustion behavior in RDE systems remains elusive. This study aims to fill this knowledge gap by analyzing the flow characteristics of a simplified RDE model using computational fluid dynamics. The findings contribute to the knowledge and understanding of RDE performance and operation, paving the way for their incorporation into next-generation propulsion.

2 Mathematical and Physical Model

The mathematical and physical model forms the backbone of any simulation study, offering insights into the complex processes occurring within the system under investigation. In this section, we examine the specific features of the simulation domain and discretization, pivotal in capturing the physics in numerical simulations of RDEs accurately.

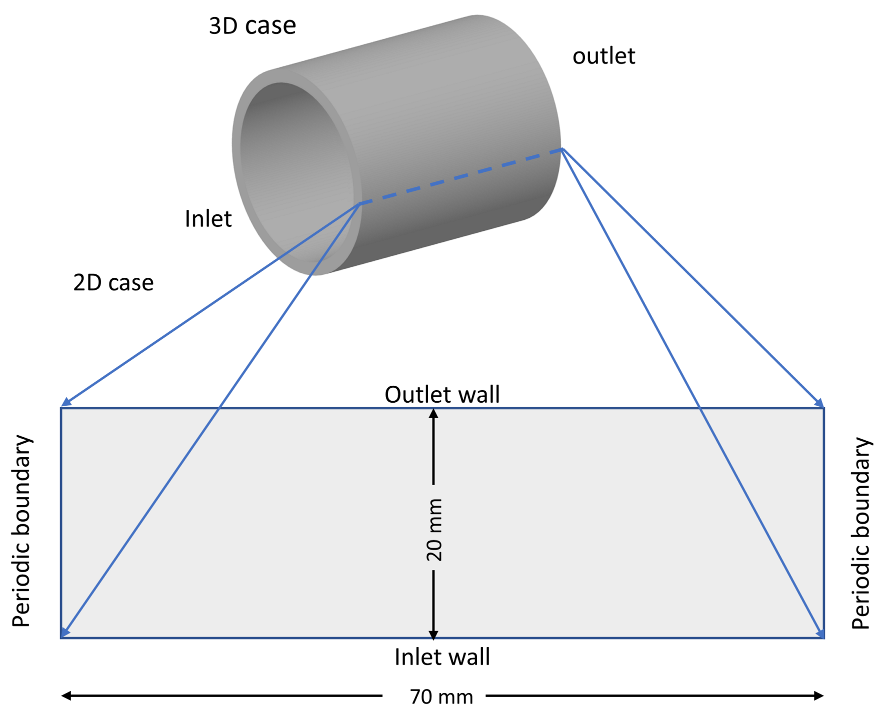

Figure 1: Two-dimensional planar geometry representing the unwrapped annular RDE chamber.

2.1 Geometry

The geometry of the computational domain is a critical factor determining the computational effort required for RDE simulations. A full 3D annular chamber model, while accurate, requires high computational costs, making it less feasible for initial studies. Therefore, a simplified planar geometry serves as a balanced choice to maintain both accuracy and efficiency.

In this study, a 2D rectangular domain measuring 20x70 mm is chosen, a decision supported by previous validations in RDE simulations [9]. This geometry, although simpler compared to the full annulus, does not compromise the integrity of the simulation, capturing the essential physics at a fraction of the computational cost. The 20 mm width of the chamber significantly reduces complexity compared to the mean diameter, allowing for the reasonable neglect of curvature effects. This simplification facilitates the modeling of the annular combustor as a planar channel, substantially reducing mesh requirements and computational cost while preserving the essential dynamics of detonation propagation.

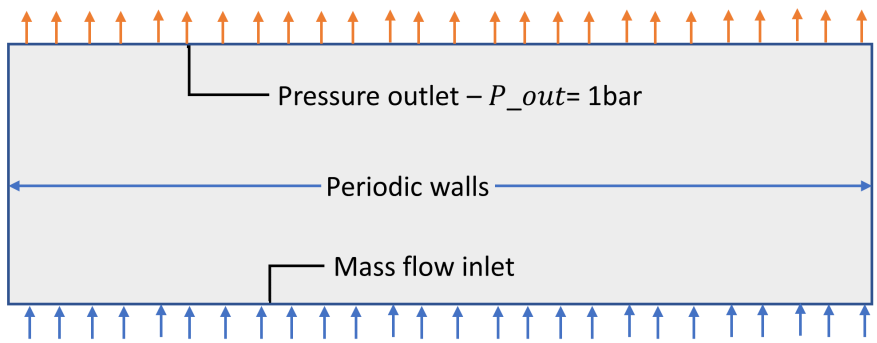

Figure 2: Two-dimensional planar geometry representing the unwrapped annular RDE chamber.

2.2 Boundary Conditions and Material Properties

Appropriate boundary conditions are necessary for obtaining realistic flow solutions in the RDE model. Mass flow inlet conditions are imposed for the H_2 and air inlets with a total mass flux of $357kg/m^2s$ and a stoichiometric equivalence ratio of 1.0. The stoichiometric equivalence ratio, a dimensionless quantity, represents the ratio of the actual fuel-to-oxidizer ratio to the stoichiometric fuel-to-oxidizer ratio. A ratio of 1.0 indicates a perfect balance, where all the fuel and oxidizer are consumed during the reaction. As depicted in Figure 2, the outlet is specified with a pressure of 1 bar and non-reflecting conditions, similar to previous studies [10]. This prevents undesired reflections and resonance in the chamber. The total temperatures are set to 300 K for both inlet and outlet.

Regarding simulation dynamics, the right and left-hand boundaries are treated as non-slip and adiabatic wall boundaries, preventing the detonation wave from propagating in both directions. Notably, during the simulation, a detonation wave emerges and traverses the domains across almost the entire width. At this point, the simulation is paused and restarted, with the wall boundary conditions replaced with translational periodic boundaries. This adjustment arises from the need to prevent the collision of the detonation wave with a shock wave traveling in the opposite direction, a phenomenon induced by the pressure difference between the left and the right wall at the end of the initial run. Consequently, the detonation wave traverses the periodic boundary, initiating propagation through the newly injected fuel/air mixture.

Table 1: Mass fractions for the stoichiometric mixture H_2/Air

Species	**Reactants(%)**	**Products(%)**
H_2	0.0287	0.2565
O_2	0.2278	0
N_2	0.7435	0.7435

Air is modeled as the oxidizer, representing an air-breathing engine as opposed to using pure oxygen, thereby simulating an engine setup found at sea level, as observed in several studies in the literature. However, modeling the true air composition significantly increases computational expenses. Therefore, following Helmenstine [11], a surrogate air mixture of 78.084% N_2 and 20.9476% O_2 is employed, providing a dilution ratio of $\beta = 3.7276$ to match realistic detonation cell sizes, a significant metric for evaluating the flow. The balanced stoichiometric reaction is:

$$2\mathrm{H}_2 + \mathrm{O}_2 + \beta\mathrm{N}_2 \rightleftharpoons 2\mathrm{H}_2\mathrm{O} + \beta\mathrm{N}_2 \tag{1}$$

The detailed CHEMKIN mechanism with 21 reactions and 10 species is implemented in ANSYS Fluent using reduced chemistry. This approach captures the coupling between detonation wave propagation and reacting flow physics effectively.

2.3 Ignition Strategy

As depicted in Figure 3, the initiation of the exothermic reaction and the subsequent detonation are achieved through the use of the direct ignition method. This method involves imposing a small, high-pressure, and high-temperature region in the lower left corner of the computational domain, measuring 0.5 mm x 5.0 mm, to initialize the detonation wave. This setup is designed to mimic a real-world application where a shock tube is attached to the combustion chamber, introducing a detonation wave into the main chamber, thereby avoiding the need for complex pre-detonator configurations. The imposed conditions of 30 bar and 3500 K within this driver section are determined based on the typical conditions generated by a shock tube, found to be sufficient to reliably initiate the propagating detonation wave [12].

This well-established combination of hydrogen-air reactants, reduced kinetic mechanism, and direct ignition approach, modeled after shock tube initiation, facilitates efficient and accurate simulations focused on analyzing the fundamental physics of rotating detonation engines.

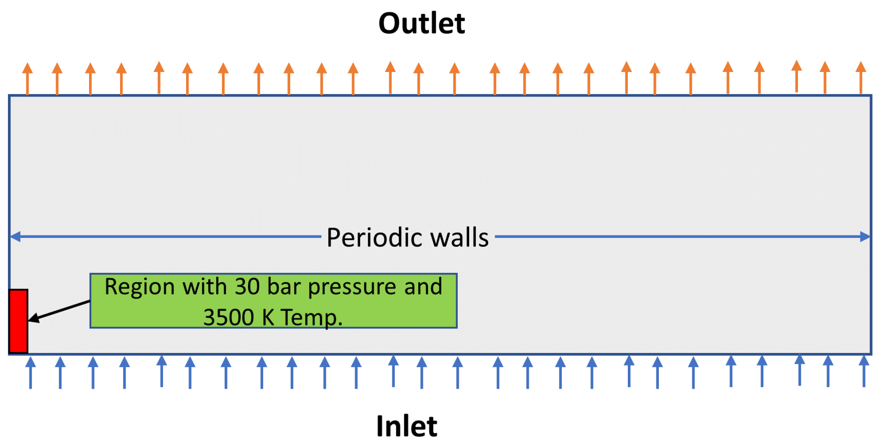

Figure 3: Direct ignition in RDE chamber.

2.4 Solver Settings and Simulation Process

The fundamental modeling approach involves the Reynolds-averaged Navier-Stokes (RANS) equations to represent conservation laws for turbulent flows, as extensively discussed in the literature [13]. To account for turbulence effects, the widely utilized $k-\omega$ SST two-equation model [14] is employed in this study. These RANS models with turbulence closures have been validated for modeling complex flows and are available in the literature [13][14].

For detonation simulations, existing finite-rate chemistry (FRC) models [15][16] are adapted, which directly compute local reaction rates without equilibrium assumptions. Species transport equations account for diffusion and reactions [15]. The Arrhenius reaction rate formulas, proportional to reactant concentrations, allow for the incorporation of detailed chemical kinetics [16].

In summary, validated RANS equations and turbulence models [13][14][16] form the base modeling approach. Adaptations of established FRC models [15][16] enable the incorporation of complex chemical mechanisms for detonation. The model selection is justified based on validation and availability in the literature.

The parameters for calculating the reaction rate coefficients for the hydrogen–oxygen mixture are given in [17]. The mechanism describes the combustion of hydrogen with oxygen, accounting for 21 reactions in a 10-component mixture. In this study, two-dimensional numerical simulation using the reduced kinetic mechanism [17] is implemented due to its availability and implementation in ANSYS Fluent [9] using CHEMKIN files.

Mechanism of hydrogen combustion with oxygen (measurement units: cm^3, mol, s, kcal, K)

No.	Reaction	A_f	β_f	E_f	A_b	β_b	E_b
1	H + O2 = O + OH	$1.92.10^{14}$	0.00	16.44	$5.48.10^{11}$	0.39	−293.28
2	O + H2 = H + OH	$5.08.10^{4}$	2.67	6.292	$2.67.10^{4}$	2.65	4.885
3	OH + H2 = H + H2O	$2.16.10^{8}$	1.51	3.43	$2.3.10^{9}$	1.4	18.338
4	O + H2O = OH + OH	$2.97.10^{6}$	2.02	13.4	$1.47.10^{5}$	2.11	−2.907
5	H2 + M = H + H + M*	$4.57.10^{19}$	−1.40	105.1	$1.15.10^{17}$	−1.68	0.82
6	O + O + M = O2 + M*	$6.17.10^{15}$	−0.50	0.00	$4.52.10^{17}$	−0.64	119
7	O + H + M = OH + M*	$4.72.10^{18}$	−1.00	0.00	$9.88.10^{17}$	−0.74	102.2
8	H + OH + M = H2O + M*	$4.50.10^{22}$	−2.00	0.00	$1.91.10^{23}$	−1.83	118.6
9	H + O2 + M = HO2 + M*	$1.48.10^{12}$	0.60	0.00	$3.48.10^{16}$	−0.41	−1.12
10	HO2 + H = H2 + O2	$1.66.10^{13}$	0.00	0.82	$3.16.10^{12}$	0.35	5.56
11	HO2 + H = OH + OH	$7.08.10^{13}$	0.00	0.30	$2.03.10^{10}$	0.72	36.88
12	HO2 + O = OH + O2	$3.25.10^{13}$	0.00	0.00	$3.25.10^{12}$	0.33	53.33
13	HO2 + OH = H2O + O2	$2.89.10^{13}$	0.00	−0.50	$5.86.10^{13}$	0.24	69.15
14	HO2 + HO2 = H2O2 + O2	$4.2.10^{14}$	0.00	11.98	$4.63.10^{16}$	−0.35	50.72
15	H2O2 + O2 = HO2 + HO2	$1.43.10^{13}$	−0.35	37.1	$1.3.10^{11}$	0.00	−1.63
16	H2O2 + M* = OH + OH + M*	$2.95.10^{17}$	0.00	48.47	$3.2.10^{4}$	0.00	45.54
17	H2O2 + H = H2O + OH	$2.41.10^{13}$	0.00	39.74	$1.27.10^{8}$	1.31	71.48
18	H2O2 + H = H2 + HO2	$6.03.10^{13}$	0.00	7.96	$1.04.10^{11}$	0.70	23.97
19	H2O2 + O = OH + HO2	$9.55.10^{6}$	2.00	0.40	$8.66.10^{3}$	2.68	18.60
20	H2O2 + OH = H2O + HO2	$1.00.10^{12}$	0.00	0.00	$1.84.10^{10}$	0.59	30.92
21	H2O2 + OH = H2O + HO2	$5.80.10^{14}$	0.00	9.57	$1.07.10^{13}$	0.59	40.48
* — reaction occurring under the action due to a third component							

The CFD modeling is implemented in ANSYS Fluent using a transient 2D approach. A second-order upwind scheme is used for spatial discretization of the convective terms. Temporal discretization is first-order implicit. The time step Δt is constrained by a Courant-Friedrichs-Lewy (CFL) condition:

$$\Delta t \leq \mathrm{CFL}\frac{\Delta x}{u} \tag{2}$$

For obtaining a stable solution, the CFL number is chosen to vary between 0.6 to 0.8, and the time-step size is 5×10^{-8} s to ensure high convergence rate.

3 Results

In the effort to understand the intricate dynamics of a rotating detonation engine (RDE), a comprehensive mesh convergence analysis was conducted, utilizing three distinct mesh configurations. The analysis was designed to decipher the implications of mesh granularity on the simulation outcomes, with a focus on the nuances of temperature predictions and shock wave dynamics.

The initial configuration, termed as Mesh 1, integrated elements with a side length of 0.2 mm. The simulations under this setup revealed a peak total temperature of 3040 K within the RDE, thereby setting a baseline for the analysis with finer mesh structures. Progressing to a more detailed level, the second configuration, Mesh 2, employed a finer mesh with elements sized at 0.1 mm, accompanied by an inflation layer in the Y-direction with a growth factor of 1.2. Further details concerning the number of layers and the minimum cell size will be integrated here to offer a deeper understanding of the mesh structure. This strategy exhibited an increase in the maximum total temperature, recording a value of 3430 K, which highlighted the influence of mesh refinement on the temperature dynamics within the engine.

Enhancing the granularity further, the third configuration, named Mesh 3, was characterized by uniform elements with a dimension of 0.1 mm, culminating in a total of 140,000 2D quadrilateral elements. Despite computational constraints limiting the grid spacing (δx) to 0.1 mm, this configuration revealed a more discernible pressure and temperature gradient. It emerged as a proficient setup for capturing the detailed dynamics of shock waves within the RDE, as evidenced by the observed peak total temperature of 3590 K.

This convergence study clearly indicates that a finer mesh resolution improves the accuracy of temperature predictions within the RDE. For the benefit of the readers, it is pertinent to highlight that the computed maximum total temperatures were 3040 K, 3430 K, and 3590 K for the Mesh 1, Mesh 2, and Mesh 3 configurations, respectively. This data emphasizes the pivotal role of mesh granularity in accurately delineating the complex dynamics of an RDE, despite the computational constraints encountered during the analysis.

The CFD simulations reveal significant insights into the complex flow behavior and combustion processes within the simplified 2D RDE model, as discussed in the following sections.

3.1 Temporal Evolution of the Detonation Wave

The analysis of pressure fluctuations, as shown in Fig 4, was conducted by placing a probe at the specified coordinates (35mm, 5mm), a location that was depicted in the preceding geometry figures to furnish a comprehensive understanding of the experimental setup. This approach illuminates the dynamic characteristics of the rotating detonation wave within the RDE, shedding light on the rhythmic recurrence of certain intrinsic phenomena within the monitored timeframe.

An important finding in this study is the observation of a stable pattern from the fourth cycle onwards, with similar pressure peaks repeating themselves. This steady pattern, shown in the figure, provides a clear view of the repeating yet complex behaviors of the wave. It suggests a stable combustion or flow process occurring in the noted cycles.

However, the limitations of the Reynolds-Averaged Navier-Stokes (RANS) method utilized in this study warrant close scrutiny. This method does not fully capture the rapid changes in turbulence in the flow, offering a more generalized view of turbulence events. Critical consideration of the chosen model is essential, as it aids in understanding the findings better and facilitates in-depth discussion in the scientific community. Moreover, the observed stability is significantly influenced by the RANS method, which does not fully capture rapid changes in turbulence, a characteristic noted in the concept of Reynolds averaging. These considerations underscore the necessity for a careful evaluation of the simulation process.

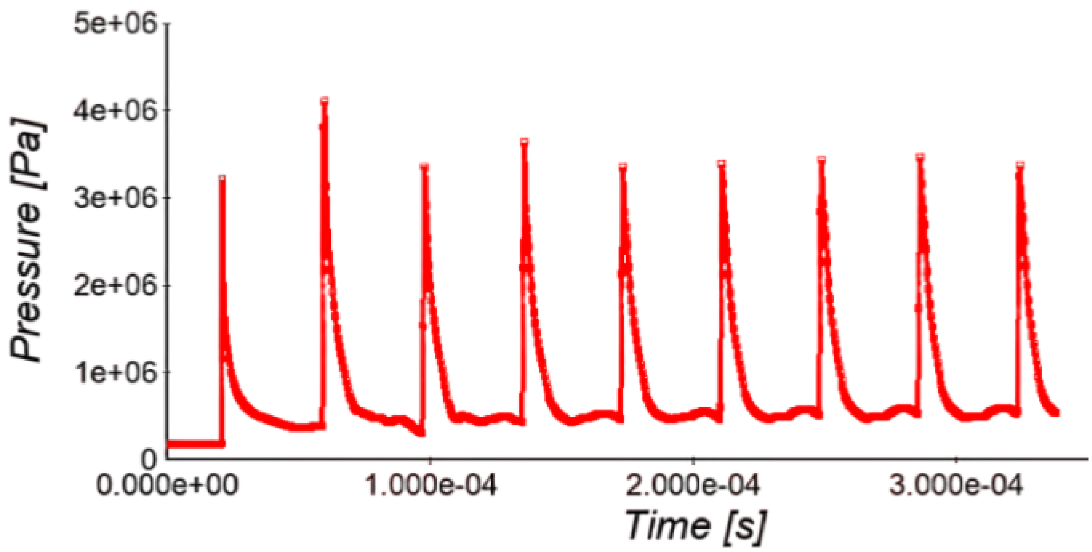

Figure 4: Time variation of pressure at the monitoring point, illustrating the consistent cycle of detonation wave propagation after initial fluctuations.

In this study, a consistent time gap of $3.75e^{-5}$ seconds was observed between successive peaks, which equated to a frequency of 80 kHz. This data denotes a wave speed of 1880 m/s, considering the chamber's 70 mm length. This computed velocity shows a slight reduction, approximately 4%, compared to the theoretical Chapman-Jouguet (CJ) velocity of 1971 m/s, expected for a hydrogen-air mixture as delineated in [**?**]. This discrepancy not only presents opportunities for optimizing the mixing process and minimizing losses associated with the curvature of the detonation front, as elucidated in [**?**], but also necessitates a detailed evaluation of the chosen numerical approach and its potential implications on the resultant interpretations.

3.2 Pressure and Temperature Distribution

The spatial distribution of both pressure and temperature within the chamber, as depicted in Fig. 5 and Fig. 6, offers comprehensive insight into the dynamics of the evolving detonation wave. These observations align significantly with the trends documented in related studies (refer to Fig. 6 and subsequent figures).

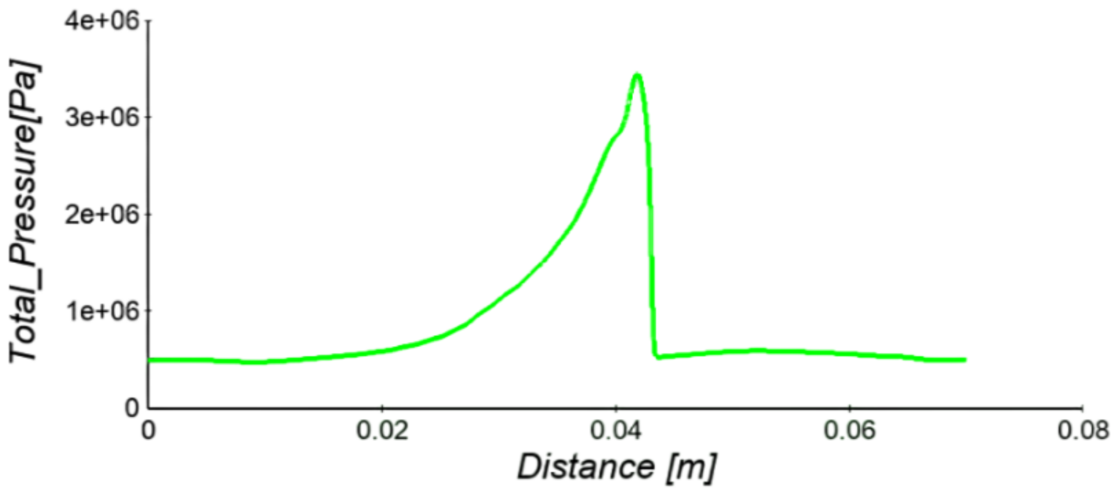

Figure 5: Pressure contour showing spike across detonation wave and subsequent expansion

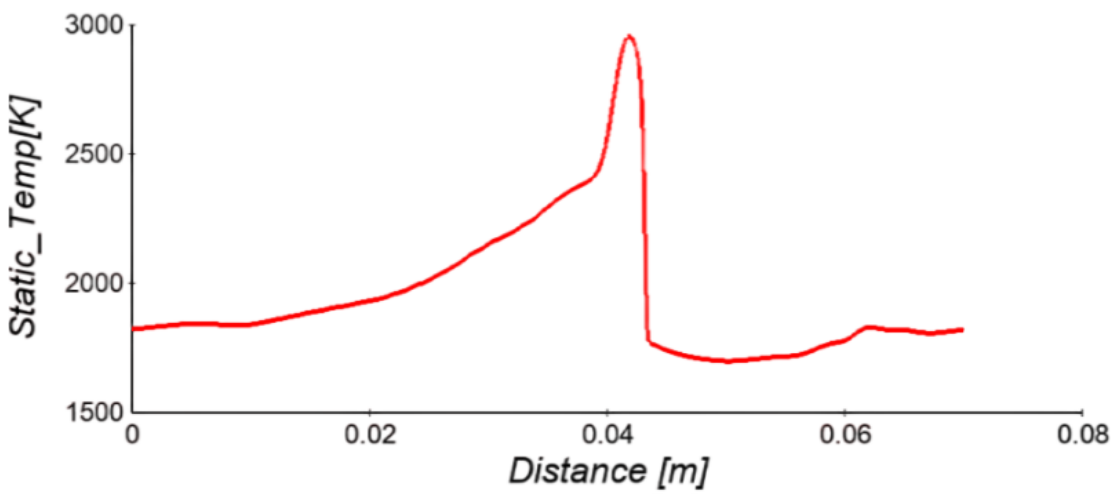

Figure 6: Temperature contour showing spike across detonation wave and subsequent expansion

In Fig. 5, the pressure profile distinctly highlights a considerable increase at the onset of the wave, marking the intense compression affecting the reactants. Following the detonation, a zone of steady and consistent pressure is observed, preventing further intake into the chamber until the expanding gases decrease the pressure to levels below the plenum conditions. This phenomenon underscores the essential role of precise timing in the injection process, which can significantly influence the efficiency of the mixing and combustion sequences.

Focusing on the temperature distribution depicted in Fig. 6, it delineates the initiation of the ignition process and the subsequent thermal propagation within the chamber. However, upon detailed scrutiny, areas exhibiting higher temperatures are apparent, hinting at potential unevenness in the mixing process at certain junctures within the chamber. These regions, clearly marked in the diagram, signal opportunities for further optimization, with the goal to foster a more uniform and efficient mixing process.

3.3 Analysis of Flow Characteristics

To gain a deeper insight into RDEs, let's visualize the flow-field from the perspective of the detonation wave. Within this context, we can trace the journey of multiple fluid segments as they traverse the combustion chamber, depicted in Fig. 7. We select trajectories that pass through various parts of the detonation wave: the lowermost section (1), the central section (2 and 3), the uppermost section (4), and a fluid segment that bypasses the detonation wave entirely (5). This phenomenon is evident in two specific regions. Initially, it's observed at the interface between the reactant and the products prior to the progression of the detonation wave. Subsequently, it's seen in the area beyond the detonation wave, where the reactant begins to seep into the combusted material.

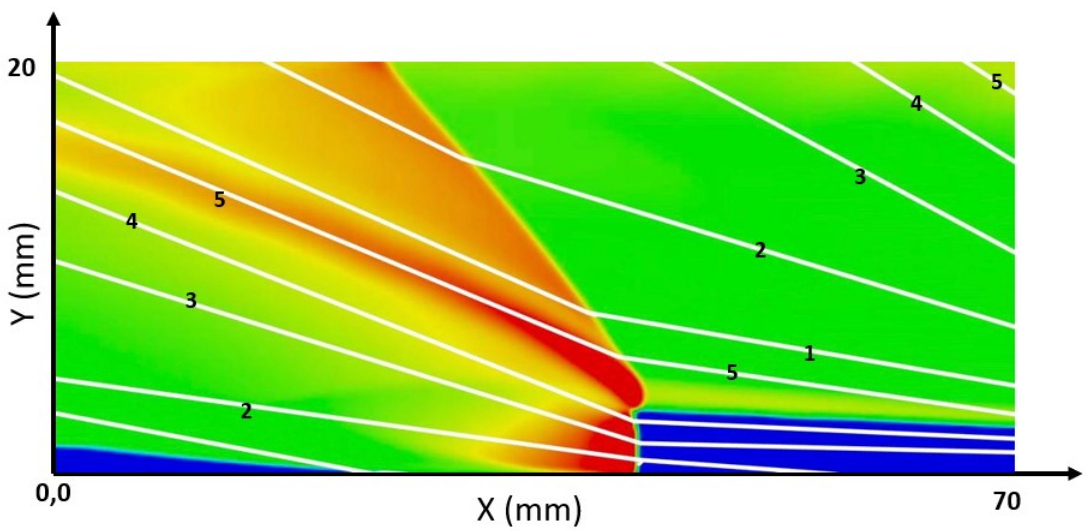

Figure 7: Variation of tangential velocity during the wave

It's crucial for the reactant to have an adequate mass flow to suppress any reactions when it intermingles with the combustion products of the detonation. Fig. 8 illustrates a Pv diagram that traces paths 1, 3, and 5 from Fig. 7. This diagram reveals that the RDE cycle primarily operates at a constant volume, with a few notable distinctions. To begin with, the routes through the lower and central parts of the detonation wave bear a close resemblance. The main deviation is observed towards the culmination of the expansion phase. For path 1, it encounters the slanting shock wave, as illustrated in Fig. 7. This wave causes a significant reduction in the expansion curve's v value. Despite this, it still follows a similar trajectory, expanding until it reaches the outlet. It's worth noting that path 3 doesn't intersect with this slanting shock wave and therefore doesn't undergo the same decrement. On the other hand, path 5 bypasses the detonation phase entirely, offering minimal contribution to the RDE's potential work output. Analyzing Fig. 8, it's evident that the thermodynamic cycle remains consistent across different flow paths within the engine. The detonation wave and the slanting shock wave are the two primary events that distinguish these paths. Considering these events, we can classify the flow paths into distinct categories. As a result, we distinguish three primary thermodynamic zones: gas that is subjected to both the detonation and the oblique shock wave, gas that only undergoes the detonation, and gas that reacts before it meets the detonation wave.

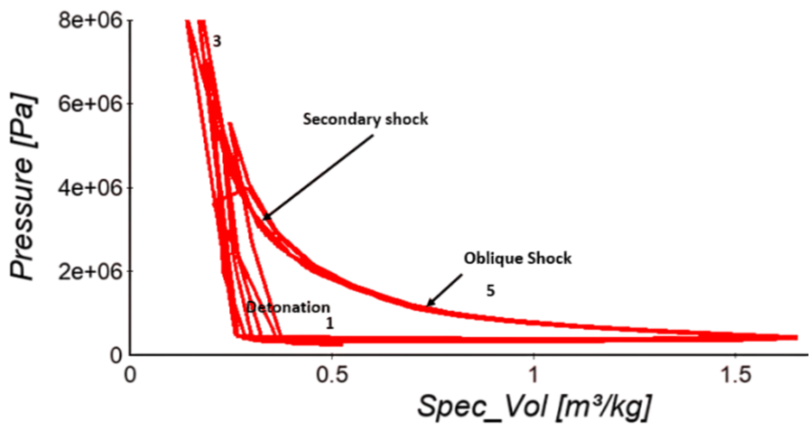

Figure 8: P-v diagram

3.4 Detonation Front Dynamics

The evolving process of the detonation wave becomes clear when examining the time-based temperature field contours. Figure 9a) indicates that the start of the detonation event occurs through the high pressure and temperature section that is set up, successfully causing the detonation as shown in Figure 9b). However, Figure 9c illustrates a brief deflagration occurring at this early stage when the new reactants mix and react with the detonation products from the initial section. The meeting point of the reacted and unreacted mixtures seems to facilitate this brief deflagration. Once the detonation wave develops fully, a smooth, even front forms, as seen in Figure 9d), marking the start of a stable cycle of operations.

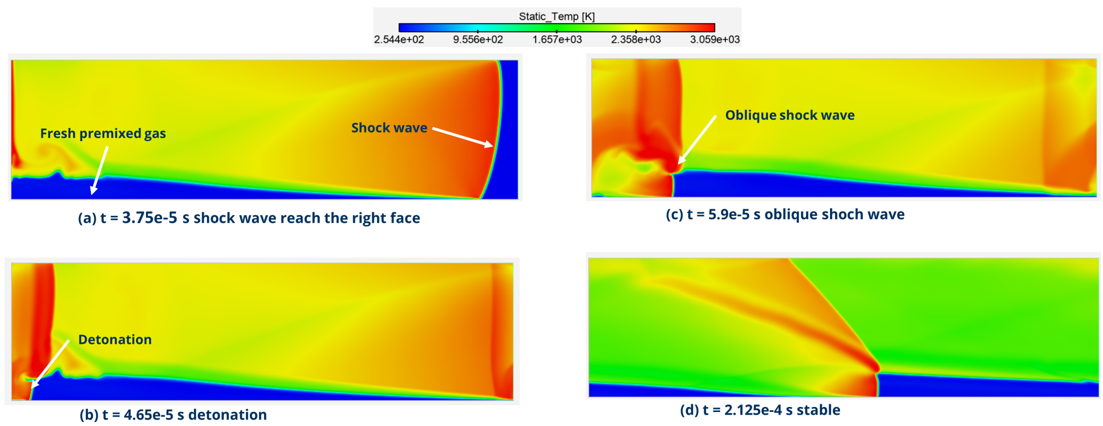

Figure 9: Temperature contour at different time steps.

The slip line, demarcating the fresh reactants and detonated products, is visible in the temperature contours as seen in Figure 11. This line marks the limit of the reaction zone, setting a clear boundary between the unburned and burned gases. This slip line plays a crucial role in understanding the progression of the reaction within the chamber, offering insights into how the fresh reactants interact with the products of detonation. Furthermore, the presence of oblique shock waves, caused by wave interactions with the chamber walls, emphasizes the necessity to implement periodic boundary conditions to avoid such unwanted effects, preserving the integrity of the observed phenomena.

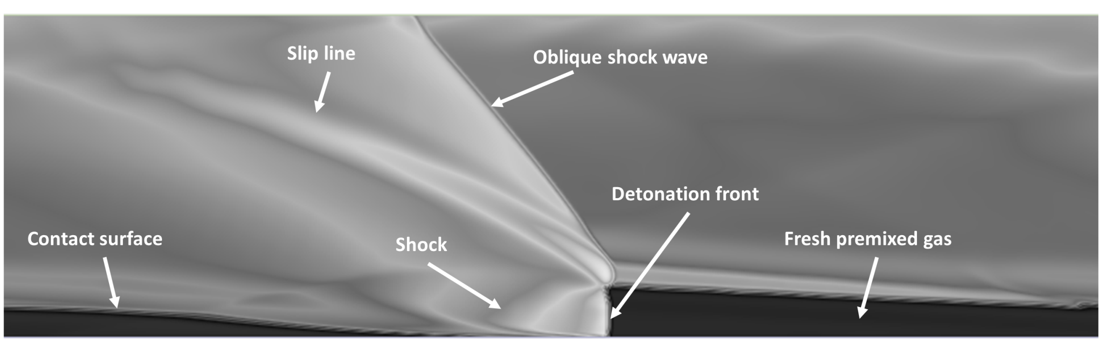

Figure 10: Flow phenomena of the traveling detonation wave

To conclude, this detailed analysis presents a thorough characterization of the complex flow and combustion dynamics governing the behavior of the rotating detonation engine system. While the findings indicate potential areas for design improvements, it is essential to note that specific strategies for enhancing mixing and minimizing losses have not been elaborated in this study. Future efforts should focus on expanding upon these identified areas. Moreover, a concise reflection on the generated model is needed in the current discussion, which would outline the capabilities and limitations of the simplified model used, offering a balanced view of the aspects that are accurately represented and those that might not be captured effectively.

4 Conclusion

This study presented a detailed investigation of the flow dynamics within a 2D model of a rotating detonation engine (RDE), aiming to improve the understanding of the intricate combustion process within such advanced propulsion systems. Several significant conclusions can be drawn from this research. One of the primary findings was the establishment of a stable detonation wave after the initial transient phase. The wave demonstrated a consistent pattern from the fourth cycle onwards, indicating the potential reliability and consistency of the RDE. The spatial distribution of pressure and temperature within the chamber provided invaluable insights into the detonation process. The observed pressure spikes and subsequent temperature rise downstream of the detonation wave underscore the release of substantial thermal energy within the RDE. However, localized temperature disparities suggest opportunities for further optimization of the mixing process. The visualization of the flow-field and the journey of fluid segments provided a comprehensive understanding of the fluid dynamics within the RDE. The presence of both the detonation and oblique shock waves played significant roles in dictating the trajectories and subsequent thermodynamics of these fluid segments. The mesh convergence analysis emphasized the pivotal role of mesh resolution in capturing the intricate dynamics of the RDE. The discrepancies between the computed and theoretical Chapman-Jouguet velocities highlighted the necessity for a meticulous evaluation of the chosen modeling approach and its impact on the resultant interpretations. The thermodynamic Pv diagram revealed that despite varying flow paths and events encountered within the chamber, the thermodynamic cycle remained predominantly consistent, operating primarily at constant volume.

In conclusion, this research stands as a significant stride in bridging the knowledge gap concerning RDEs. By unraveling the complexities of the flow behavior and detonation dynamics within a simplified RDE model, this study illuminates the path for further advancements in RDE design and optimization. The insights garnered here contribute to both the academic understanding and applied advancement of rotating detonation engines.

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